

can

principled way to extract smooth approximate maps

Barycentric Projection

- But for Entropic OT, we **can** define an out-of-sample extension
[Seguy et al 2018; Pooladian & Niles-Weed 2021]
- Define the entropic transport map as:

$$\text{For any } x \in \mathcal{X} : \quad T_\varepsilon(x) = \frac{\int y \exp\left(\frac{1}{\varepsilon}(g_\varepsilon(y) - c(x, y))\right) d\beta(y)}{\int \exp\left(\frac{1}{\varepsilon}(g_\varepsilon(y) - c(x, y))\right) d\beta(y)}.$$

- Where does this come from?
 - Pooladian & Niles-Weed (2021): $T_\varepsilon = \text{Id} - \nabla f_\varepsilon$ where f_ε is the optimal dual entropic potential.
 - Interpreted as the expected target location under the Gibbs distribution defined by g_ε and $c(x, y)$.
- **Takeaway:** Entropic regularization not only gives us efficient couplings, but also a **principled way to extract smooth approximate maps**.

Barycentric Projection: Application

Optimal Transport for Domain Adaptation

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Abstract—Domain adaptation is one of the most challenging tasks of modern data analytics. If the adaptation is done correctly, models built on a specific data representation become more robust when confronted to data depicting the same classes, but

This transformation can be conveniently expressed with respect to the target samples as the following **barycentric mapping**:

$$\widehat{\mathbf{x}}_i^s = \operatorname{argmin}_{\mathbf{x} \in \mathbb{R}^d} \sum_j \gamma_0(i, j) c(\mathbf{x}, \mathbf{x}_j^t). \quad (13)$$

where $\mathbf{x}_i^s$ is a given source sample and $\widehat{\mathbf{x}}_i^s$ is its corresponding image. When the cost function is the squared ℓ_2 distance, this barycenter corresponds to a weighted average and the sample is mapped into the convex hull of the target samples. For all source samples, this barycentric mapping can therefore be expressed as:

$$\hat{\mathbf{X}}_s = \mathbf{T}_{\gamma_0}(\mathbf{X}_s) = \operatorname{diag}(\gamma_0 \mathbf{1}_{n_t})^{-1} \gamma_0 \mathbf{X}_t. \quad (14)$$

